Fuel 429 (2027) 140722



Contents lists available at ScienceDirect

Fuel

journal homepage: www.elsevier.com/locate/jfue



Full Length Article

Thermochemical pre-combustion fuel transformation as “exergy pump”: a case of ammonia for power generation

Dmitry Pashchenko [a, b]

[a] Faculty of Mechanical Engineering, Technion - Israel Institute of Technology, Technion City, Haifa 3200003, Israel
[b] Department of Mechanical Engineering (Robotics) and MATEC, Guangdong Technion–Israel Institute of Technology, 241 Daxue Road, Guangdong, Shantou, 515063, China

HIGHLIGHTS

- “Exergy pump” upgrades low-grade heat via endothermic fuel reforming.
- Ammonia decomposition at 700 °C boosts input heat exergy by 27.5%.
- Solar ammonia reforming at 500 °C raises cycle efficiency to 50.52%.
- Enables efficient use of medium-temperature heat in high-efficiency cycles.

ARTICLE INFO

Keywords:
Ammonia
Gas turbine
Exergy
Thermal efficiency
Combined cycle plant

ABSTRACT

This paper introduces and rigorously defines the concept of an “exergy pump” – a thermochemical strategy that upgrades low- and medium-grade heat by using it to drive endothermic pre-combustion fuel reforming. Using ammonia as a model carbon-free fuel, this study demonstrates how its endothermic decomposition serves as an effective pre-combustion reforming process for energy quality enhancement. The objective of this work is to conduct a thermodynamic analysis of the concept for both an idealized reference case (Carnot cycle) and a realistic power generation system (combined cycle). The methodology is based on exergy analysis, which enables assessment of energy quality at different stages of the process. Novel performance indicators are introduced and quantified: the increase in fuel exergy, the coefficient of heat transformation (CHT), and the coefficient of exergy transformation (CET). For the Carnot cycle reference case, it is shown that supplying medium-temperature heat at 700 °C for ammonia decomposition increases the exergy of this heat stream by 27.5%. For a combined cycle power plant based on the Mitsubishi H-25 gas turbine, it is demonstrated that utilizing solar thermal energy at 500 °C for pre-combustion ammonia decomposition increases the overall cycle efficiency from 46.18% to 50.52%. From an exergy perspective, this configuration is 23.9% more effective compared to the direct use of the same solar heat for steam generation. The proposed approach additionally provides a pathway for the efficient utilization of low- and medium-temperature heat streams in high-efficiency gas turbines, including those operating on promising carbon-free fuels.

1. Introduction

The global energy landscape is undergoing a profound transformation, driven by the dual imperatives of ensuring energy security and mitigating climate change. While renewable energy sources are experiencing rapid growth, thermal power generation – particularly from gas turbine and combined cycle plants – remains the backbone of the global electricity supply, providing essential stability and dispatchability to the grid [1,2]. This enduring reliance creates a complex set of interrelated challenges for modern thermal power engineering. Consequently, improving the efficiency of these systems and reducing their carbon footprint is not merely an option but a critical necessity for a sustainable energy future [3].

Traditional efforts to enhance the performance of fuel-converting technologies have historically concentrated on maximizing the first-law efficiency, which quantifies the quantity of energy converted. However, this perspective overlooks the quality, or “usefulness”, of energy, a concept rigorously defined by the second law of thermodynamics through exergy [4]. Exergy represents the maximum useful work obtainable from a system as it reversibly comes into equilibrium with its environment.

Email addresses: dmitry.pa@technion.ac.il; dmitry.pashchenko@gtiit.edu.cn.

https://doi.org/10.1016/j.fuel.2026.140722
Received 27 April 2026; Received in revised form 30 June 2026; Accepted 15 July 2026
Available online 21 July 2026
0016-2361/© 2026 The Author. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Nomenclature

Abbreviations

CCGT	Combined Cycle Gas Turbine
CET	Coefficient of Exergy Transformation
CHT	Coefficient of Heat Transformation
CSP	Concentrating Solar Power
GAH	Gross Available Heat
HRSG	Heat Recovery Steam Generator
LHV	Lower Heating Value
TCR	Thermochemical Reforming
TIT	Turbine Inlet Temperature
SSG	Solar Steam Generator

Symbols

Ex	Exergy (MJ/kg or MW)
Ex_{fuel}	Chemical exergy of fuel (MJ/kg)
Ex_{prod}	Chemical exergy of reaction products (MJ/kg)
Ex_q	Thermal exergy (MJ/kg or MW)
ΔEx_{fuel}	Increase in fuel exergy (MJ/kg)
H^{comb}	Gross available heat (MJ/kg)
H^{in}	Sensible enthalpy of fuel (MJ/kg)
ΔH_{298}	Standard reaction enthalpy at 298 K (kJ/mol or MJ/kg)
ΔH_{gas}	Sensible enthalpy of gas mixture (MJ/kg)
ΔH_{react}	Total heat input for reaction (MJ/kg)
ΔH_{sol}	Solar heat input (MW)
$k_{\text{ex,q}}$	Coefficient of exergy transformation (–)
$k_{\text{q(I)}}$	Coefficient of heat transformation based on LHV (–)
$k_{\text{q(II)}}$	Coefficient of heat transformation based on GAH (–)
LHV	Lower heating value (MJ/kg)
P	Pressure (bar or kPa)
P_0	Reference (ambient) pressure (kPa)
Q	Heat stream (MJ/kg or MW)
Q_{in}	External heat input (MJ/kg)
T	Temperature (K or °C)
T_0	Reference (ambient) temperature (K)
X_{NH_3}	Ammonia conversion fraction (–)

Greek symbols

λ	Excess air ratio (–)

Subscripts

0	Reference (ambient) condition
fuel	Primary fuel (ammonia)
prod	Reaction products (H_2-rich gas)
q	Thermal (heat)
sol	Solar

Unlike energy, exergy is consumed in all real, irreversible processes. Therefore, minimizing exergy destruction is the true hallmark of a thermodynamically perfect system, and any strategy that can conserve or, remarkably, increase the exergy of an energy carrier is of immense scientific and practical interest [5].

A major source of inefficiency and exergy loss in modern thermal systems is the under utilization of low- and medium-grade heat. Vast quantities of low- and medium-temperature thermal energy (in the range of 200–700 °C) are rejected to the environment from industrial processes and power plants. While abundant, this energy is of low quality. According to the Carnot principle, the maximum theoretical efficiency for converting this heat into work is severely limited by its relatively low temperature [6]. The direct application of this heat in bottoming cycles, such as steam Rankine cycles or Organic Rankine cycles, often results in low exergy efficiencies due to the moderately low temperature of working fluid. The fundamental challenge, therefore, is not just to recover this heat, but to upgrade its quality – to increase its exergy content – without violating the second law, which prohibits the spontaneous transfer of heat from a colder body to a hotter one [7].

This is where the concept of an “exergy pump” emerges as a transformative energy management strategy. The core idea is to use low- or medium-temperature heat to drive an endothermic reaction of fuel reforming before combustion, thereby converting this low-quality thermal energy into high-quality chemical energy. In this process, the reaction acts as a chemical “compressor” or “pump”, absorbing heat at a moderate temperature and “lifting” the energy to a higher potential by storing it in the chemical bonds of the reaction products [8]. When these products are subsequently combusted, the energy is released at a higher flame temperature, effectively realizing the upgraded exergy content [9,10]. This approach represents a paradigm shift from passive heat recovery to active energy upgrading, with profound implications for the design of future power systems [11,12].

In this study, the thermochemical pre-combustion transformation of fuel is analyzed as an “exergy” pump, using the endothermic decomposition of ammonia (NH_3) as a model thermochemical process. Ammonia has recently emerged as a leading candidate for a carbon-free energy carrier due to its high hydrogen density and well-established infrastructure for storage and transport [13–16]. However, the direct utilization of ammonia in combustion systems is hampered by unfavorable combustion kinetics, including a low flame speed, high ignition temperature, and narrow flammability limits. These characteristics pose significant challenges for stable and efficient operation in devices such as gas turbines [17,18]. The thermochemical conversion of ammonia into a hydrogen-rich mixture ($3\,H_2 + N_2$) elegantly addresses both challenges simultaneously: it utilizes low-temperature heat to produce a superior fuel with better combustion properties, thereby increasing the overall thermal and combustion efficiency of the system [19].

While the potential benefits of such a concept are significant, a rigorous thermodynamic assessment is required to quantify its viability. Existing literature has explored ammonia decomposition and thermochemical recuperation, but a focused exergy analysis that frames this process explicitly as a tool for energy quality upgrading—an “exergy pump”—is less developed, particularly in the context of its integration into real-world power cycles like combined cycle gas turbine (CCGT) plants. This study aims to fill this gap.

The novelty of this work is threefold:

- Concept and metrics: Introduction of the “exergy pump” concept for pre-combustion fuel transformation, together with three novel performance indicators — the increase in fuel exergy (ΔEx_{fuel}), the coefficient of heat transformation (CHT, k_{q}), and the coefficient of exergy transformation (CET, $k_{\text{ex,q}}$).
- First application to ammonia: Systematic exergy-based evaluation of ammonia as a carbon-free fuel for exergy pumping, including complete thermodynamic analysis of its decomposition behavior under realistic gas turbine operating pressures.
- Practical validation: Quantitative assessment of the concept integrated into an H-25-based CCGT plant (80 MW), demonstrating that using solar heat at 500 °C for ammonia decomposition improves cycle efficiency by 23.9% compared to conventional solar steam generation.

The specific objectives of this study are: (i) to conduct a detailed thermodynamic and exergy analysis, establishing key performance indicators such as the exergy upgrade ratio; (ii) to quantify the energy and exergy coefficients of transformation, demonstrating the fundamental principle of exergy amplification; and (iii) to validate the practical

potential of the concept through a case study of an integrated solar-assisted combined cycle gas turbine plant fueled with pre-combustion ammonia decomposition using solar heat, comparing its performance against a conventional configuration where the same solar heat is used directly for steam generation. By doing so, this work provides a clear, quantitative foundation for fuel-flexible power generation systems that can actively manage and upgrade energy quality, contributing directly to the goals of decarbonization and enhanced resource utilization.

2. Methodology

In this study, the term "exergy pump" is introduced to describe a thermochemical strategy that upgrades low- or medium-grade thermal energy into higher-quality chemical exergy. The concept operates by using an external heat stream (e.g., solar heat or exhaust gas) to drive an endothermic pre-combustion fuel reforming reaction. The reaction absorbs heat at a moderate temperature and stores it as additional chemical energy in the reformed fuel. Upon subsequent combustion, this stored energy is released at a higher flame temperature, effectively "pumping" the exergy content of the original heat stream to a higher level. Thus, unlike a conventional heat pump that transfers thermal energy against a temperature gradient, an "exergy pump" transforms low-value heat into a more valuable form of energy – chemical exergy – without violating the second law of thermodynamics. In the present work, the endothermic decomposition of ammonia (NH_3) into hydrogen and nitrogen serves as a practical realization of this concept.

2.1. Exergy analysis

The study of the "exergy pump" concept is based on exergy analysis, a method that allows for the assessment of energy quality. The exergy of a gaseous fuel (Ex_{fuel}) is determined from an exact chemical exergy calculation based on standard thermodynamic data:

$$Ex_{fuel} = \sum_i y_i \cdot Ex_{ch,i} + \overline{R}T_0 \sum_i y_i \ln y_i, \quad (1)$$

where $Ex_{ch,i}$ is the standard chemical exergy of component i, and the second term accounts for mixing exergy.

The thermal exergy (Ex_q) of a heat stream (Q) at temperature (T) is calculated using the following formula:

$$Ex_q = Q\left(1 - \frac{T_0}{T}\right), \quad (2)$$

where Q is the heat stream at temperature T, and T_0 is the reference (ambient) temperature. The term $(1 - T_0/T)$ is the Carnot factor, which represents the maximum fraction of heat that can be converted into work between two temperature levels.

To analyze the concept of the "exergy pump", a steady-state process is considered within a system consisting of three main units:

1. A pre-combustion thermochemical fuel reforming reactor, which receives primary fuel with a lower heating value (LHV_{fuel}) and a heat stream (Q_{in}) from an external source to compensate for the endothermic effect of the fuel transformation reaction;
2. An external heat source supplying Q_{in} at temperature T;
3. A combustion chamber for the useful utilization of the reformed fuel, where the resulting mixture is burned.

The following indicators are introduced as criteria for assessing the performance of thermochemical pre-combustion fuel transformation as an "exergy pump":

- Increase in fuel exergy, determined as the difference between the exergy of the transformation products and the exergy of the primary fuel:

$$\Delta Ex_{fuel} = Ex_{prod} - Ex_{fuel}. \quad (3)$$

- Coefficient of heat transformation (CHT): Two types of CHT are introduced. The first, $k_{q(I)}$, is the ratio of the lower heating value (LHV) of the product gas to the LHV of the initial fuel. The second, $k_{q(II)}$, is the ratio of the gross available heat obtained from the products of fuel transformation to the gross available heat of the initial ammonia:

$$k_{q(I)} = \frac{LHV_{prod}}{LHV_{fuel}}; \qquad k_{q(II)} = \frac{Q^{comb}_{prod}}{Q^{comb}_{fuel}}. \quad (4)$$

In Eq. (4), the gross available heat (GAH) Q_i^{comb} is defined as the total heat released during the complete combustion of a fuel. It is calculated as the sum of the fuel's lower heating value and its sensible heat:

$$Q^{comb}_{fuel} = LHV_{fuel} + Q^{in}_{fuel}, \quad (5)$$

where Q^{in}_{fuel} is the sensible enthalpy of the primary fuel before combustion.

- Coefficient of exergy transformation (CET) for pre-combustion endothermic fuel reforming, defined as the ratio of the increase in exergy of the fuel stream to the exergy of the supplied heat stream:

$$k_{ex,q} = \frac{\Delta Ex_{fuel}}{Ex_q}. \quad (6)$$

Throughout this study, the coefficient of exergy transformation (CET, $k_{ex,q}$) is defined as the ratio of the increase in chemical exergy of the fuel stream ($\Delta Ex_{fuel} = Ex_{prod} - Ex_{fuel}$) to the thermal exergy of the supplied heat (Ex_q). This is the definition given in Eq. (11) and is used consistently in all quantitative assessments. However, in the Carnot cycle analysis presented in Section 3.2, the exergy of combustion products is expressed in terms of thermal exergy based on the Carnot factor, because the goal there is to illustrate the final work potential after combustion. This does not change the definition of CET. Rather, it demonstrates that the chemical exergy gain (ΔEx_{fuel}) ultimately translates into a higher flame temperature and thus higher thermal exergy of the combustion products. To avoid confusion, the following distinction is made throughout the revised manuscript: CET always refers to the chemical exergy increase of the fuel divided by the thermal exergy of the input heat. When thermal exergy of combustion products is discussed, it is explicitly stated as such (e.g., $Ex_{q,comb}$). The two concepts are linked by the combustion process, but they are not interchangeable. This clarification ensures consistent use of terminology and aligns with standard exergy analysis practice.

2.2. Initial parameters

As introduced previously, the concept of an "exergy pump" applies when thermochemical pre-combustion transformation occurs. Many hydrocarbon and carbon-free fuels can be endothermically reformed prior to combustion [20,21]. In this study, ammonia is used as a model fuel due to the significant interest it has recently garnered among engineers and researchers as a potential carbon-free fuel for both existing and newly designed power plants [22–24].

The thermochemical decomposition of ammonia to produce a hydrogen-containing gas with a hydrogen volume fraction of up to 75% is a well-studied process and is described by the following equation [25,26]:

$$2NH_3 \leftrightarrow 3H_2 + N_2, \quad \Delta H_{298} = 46.03\ \mathrm{kJ/mol_{NH_3}}. \quad (7)$$

Reaction (7) is endothermic, and both the ammonia conversion and the resulting product gas composition depend on technological and design parameters. The product of this reaction is a hydrogen-rich gas.

The lower heating values of ammonia and hydrogen were taken as 18.6 MJ/kg and 120.0 MJ/kg, respectively. To determine the product gas composition and the heat required for ammonia decomposition, a thermodynamic analysis was performed.

The standard state of the environment is assumed: $T_0 = 298.15$ K, $P_0 = 101.3$ kPa.

The present analysis assumes stoichiometric combustion ($\lambda = 1.0$) for calculating adiabatic flame temperatures and thermal exergy; variations in excess air ratio would alter flame temperatures and Carnot factors, and a sensitivity analysis of λ in the range 0.9–1.5 is recommended for future work tailored to specific combustor designs.

2.3. System boundaries and interpretation of exergy increase

To avoid ambiguity in the interpretation of the "exergy pump" concept, the system boundaries are defined as follows. The system includes three main components: (i) an endothermic pre-combustion fuel reformer, (ii) an external heat source supplying thermal energy at temperature T, and (iii) a combustion chamber where the reformed fuel is burned. The primary fuel (ammonia) enters the system with a baseline chemical exergy Ex_{fuel}. The external heat stream Q_{in} enters the system with thermal exergy Ex_q calculated via Eq. (10). Inside the reformer, the endothermic reaction converts part of the supplied heat into additional chemical exergy stored in the reformed fuel. Thus, the fuel exergy increases from Ex_{fuel} (inlet) to Ex_{prod} (outlet). The increase $\Delta Ex_{\text{fuel}} = Ex_{\text{prod}} - Ex_{\text{fuel}}$ is strictly the gain in chemical exergy of the fuel stream attributable to the external heat; it does not represent a net exergy increase of the whole system because the exergy of the external heat stream is consumed in the process. The coefficient of exergy transformation (CET, $k_{\text{ex,q}} = \Delta Ex_{\text{fuel}}/Ex_q$) can exceed unity, meaning that the chemical exergy gained by the fuel is larger than the thermal exergy of the supplied heat. This is possible because the baseline exergy of the primary fuel serves as a "carrier" that enables the exergy upgrade, analogous to a heat pump where work input transfers more thermal energy than the work consumed. However, in the exergy pump, the upgraded exergy is realized only after combustion, when the reformed fuel releases heat at a higher flame temperature. The analysis focuses on the exergy contribution of the external heat stream and does not claim a violation of exergy conservation for the closed system (heat source + fuel + environment). This clarification ensures that the reported exergy enhancement is properly understood within the defined system boundaries.

3. Results and discussion

3.1. Thermodynamics of ammonia decomposition

Thermodynamic analysis allows for establishing the influence of temperature and pressure on the degree of ammonia conversion and the heat balance of the thermochemical ammonia decomposition process. Based on the Gibbs free energy minimization method, the equilibrium composition and heat balance components for reaction (7) were determined. The thermodynamic analysis methodology is described in the author's previous works [27,28].

Ammonia decomposition is an endothermic process in which the total number of moles of products is greater than the number of moles of initial ammonia. According to Le Chatelier's principle, the equilibrium gas composition is therefore dependent on both temperature and pressure. To assess the effect of temperature and pressure on the material and heat balance of the ammonia decomposition process, the molar fractions, ammonia conversion, and heat balance outputs were determined. Fig. 1 shows (a) the molar fractions of the decomposition products, (b) the ammonia conversion (X_{NH_3}), and (c) the heat absorbed by the reaction, which includes both the endothermic reaction enthalpy (ΔH_{298}) and the sensible enthalpy increase of the reacting mixture (ΔH_{gas}). The nominal pressure ratio of the Mitsubishi H-25 gas turbine is 17.9. Consequently, the molar fractions and heat balance were calculated for this pressure of 17.9 bar.

Fig. 1(a) illustrates the progressive decomposition of ammonia with increasing temperature. As temperature rises, the NH_3 fraction declines, while the products N_2 and H_2 increase, approaching near-complete decomposition above approximately 600 °C. This behavior confirms the expected equilibrium shift of the endothermic reaction at elevated temperatures, as verified by experiments [29].

Fig. 1(b) shows the ammonia conversion (X_{NH_3}) as a function of temperature for two pressures: 10 bar and 20 bar. The results clearly demonstrate the inhibiting effect of increased pressure on conversion, consistent with Le Chatelier's principle for a reaction that increases the number of gas moles. At 10 bar, high conversion (> 90%) is achieved above 600 °C, whereas at 20 bar, conversion is lower in the same temperature range.

The enthalpy analysis in Fig. 1(c) distinguishes between the reaction enthalpy (ΔH_{298}) and the temperature-dependent sensible enthalpy of the gas mixture (ΔH_{gas}). The reaction enthalpy represents the fixed endothermic heat demand for decomposing ammonia at reference conditions. For TCR design, the total enthalpy input – the sum of both contributions – must be supplied by the available exhaust heat stream.

It should be noted that the data presented in Fig. 1 correspond to the thermodynamic equilibrium state. However, sufficient information is available in the literature indicating that a temperature range of 550–600 °C is favorable for achieving a gas composition close to equilibrium when using industrial Ni-based catalysts [29,30].

3.2. Quantitative assessment

The initial data for the quantitative assessment are as follows:

- Ammonia decomposition temperature (T): 973 K
- Heat input for ammonia decomposition (ΔH_{react}): 4.99 MJ/kg$_{\text{NH}_3}$
- Lower heating value of ammonia (LHV): 18.60 MJ/kg

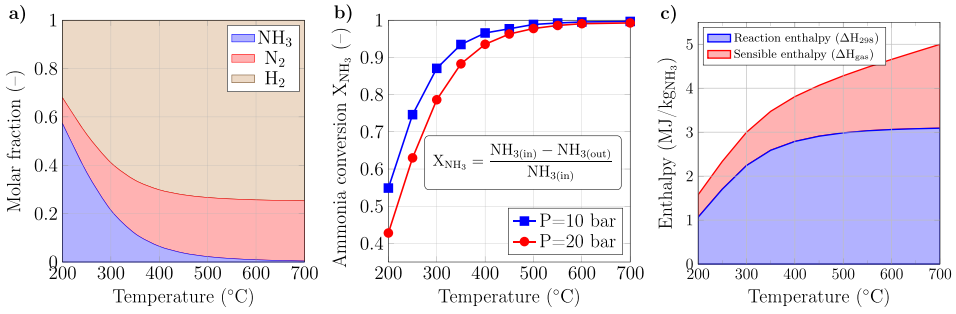


Fig. 1. Ammonia decomposition analysis: (a) molar fraction of NH_3, N_2, and H_2 versus temperature at 20 bar; (b) ammonia conversion versus temperature at 10 and 20 bar; (c) reaction enthalpy and sensible enthalpy versus temperature at 17.9 bar.

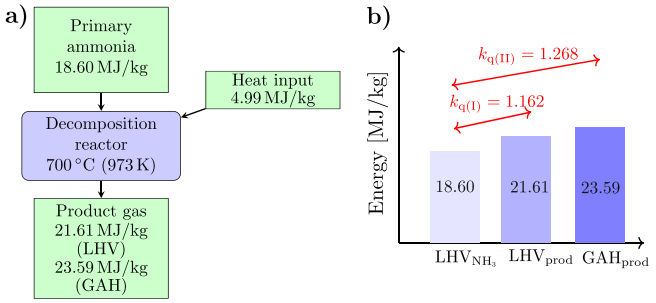


Fig. 2. Quantitative assessment of energy balance: a) energy flow diagram showing inputs and outputs; b) comparison of energy values with corresponding heat transformation coefficients.

For a quantitative assessment of the efficiency of endothermic ammonia conversion as an "exergy pump", consider the following example. The initial ammonia has a gross available heat equal to its lower heating value of 18.6 MJ/kg. To decompose 1 kg of ammonia at a temperature of 700 °C, 4.99 MJ must be supplied (1.98 MJ for heating the reaction mixture as sensible enthalpy and 3.01 MJ to cover the endothermic effect of reaction (7)). Consequently, the gross available heat obtained from the combustion of the ammonia decomposition products is 23.59 MJ/kg.

Based on the results of thermodynamic analysis, the energy balance of pre-combustion fuel transformation was compiled and summarized in Fig. 2. The coefficients of heat transformation $k_{q(I)}$ and $k_{q(II)}$ were determined as follows:

$$k_{q(I)} = \frac{21.61\,\text{MJ/kg}}{18.60\,\text{MJ/kg}} = 1.162; \qquad k_{q(II)} = \frac{23.59\,\text{MJ/kg}}{18.60\,\text{MJ/kg}} = 1.268. \tag{8}$$

The coefficient of heat transformation $k_{q(I)} = 1.162$ indicates that the lower heating value of the hydrogen-rich gas (product fuel) is 1.162 times higher than that of the primary ammonia. According to the energy conservation equation, this means that 3.01 MJ of heat at a temperature of 700 °C was converted into 3.01 MJ of chemical energy in the new product fuel. Considering that after ammonia transformation the product is at a temperature of 700 °C, the coefficient $k_{q(II)} = 1.268$ shows that the combustion of the ammonia decomposition products generates 1.268 times more heat than the combustion of primary ammonia.

Fig. 2 provides an overview of the energy balance of the pre-combustion fuel reforming system. However, heat is a quantitative indicator of energy efficiency, while exergy also considers its quality. Therefore, to better understand thermochemical pre-combustion fuel transformation as an "exergy pump", an exergy balance was compiled.

There are two energy streams at the inlet to the decomposition reactor. The first (1) is the chemical energy of ammonia, expressed in the form of its lower heating value, and the second (2) is a heat stream for thermochemical ammonia decomposition at a temperature of 700 °C. These energy streams amount to 18.60 MJ and 4.99 MJ per 1 kg of ammonia, respectively.

If these energy streams were used separately in a Carnot cycle with a sink temperature of 25 °C, the maximum possible efficiency of heat-to-work conversion, or thermal exergy, can be expressed via (10). For the exergy analysis, it is assumed that ammonia is burned with an excess air ratio of 1.0 ($\lambda = 1.0$). The adiabatic flame temperature of stoichiometric ammonia combustion in air is approximately 1800 °C [31,32]. Therefore, it is assumed that a Carnot cycle operates between temperatures of 1800 °C and 25 °C. The second stream is a heat input with temperature of 700 °C, and the hypothetical Carnot cycle for this stream operates between 700 °C and 25 °C. Considering the results of the thermodynamic analysis (Fig. 1) and energy balance (Fig. 2), the following results are obtained:

$$Ex_{q,1} = 18.60\,\frac{\text{MJ}}{\text{kg}}\left(1 - \frac{298\,\text{K}}{2073\,\text{K}}\right) = 15.93\,\frac{\text{MJ}}{\text{kg}};\; Ex_{q,2} = 4.99 \times \frac{MJ}{kg}\left(1 - \frac{298\,K}{973\,K}\right) = 3.46\,\frac{MJ}{kg}. \tag{9}$$

The results show that the maximum possible efficiency of heat-to-work conversion for the first stream is 0.856 and for the second stream is 0.694. However, it should be noted that the actual efficiency of real power cycles, e.g., Brayton or Rankine, is significantly lower (30–40%). The total exergy of the two streams is (15.93 + 3.46) = 19.39 MJ/kg.

When thermochemical pre-combustion ammonia decomposition is applied, the heat conversion occurs in a different manner. The second heat stream is absorbed by the first stream of ammonia and realized in the chemical energy and sensible enthalpy of ammonia decomposition products – an H_2-rich gas. The combustion of this H_2-rich gas generates a third stream that contains the same amount of thermal energy as the sum of the energy of the two inlet streams. The combustion of this gas with $\lambda = 1.0$ provides an adiabatic temperature of approximately 2300 °C. Thus, the 4.99 MJ of heat supplied to the decomposition reactor per 1 kg of ammonia, after combustion of its thermochemical transformation products, is expressed as heat of combustion products with a temperature of approximately 2300 °C. Therefore, the exergy of the third heat stream can be calculated as follows:

$$Ex_{q,3} = \left(18.60\,\frac{\text{MJ}}{\text{kg}} + 4.99\,\frac{\text{MJ}}{\text{kg}}\right)\left(1 - \frac{298\,\text{K}}{2573\,\text{K}}\right) = 20.85\,\frac{\text{MJ}}{\text{kg}}. \tag{10}$$

When the products of thermochemical ammonia decomposition are used as fuel, the maximum theoretical efficiency for a Carnot cycle operating between 2300 °C and 25 °C is (1 – 298/2573) = 0.884. Therefore, it can be concluded that both streams (LHV of ammonia of 18.60 MJ and external heat of 4.99 MJ) after ammonia decomposition and combustion of H_2-rich gas have a thermal exergy of 20.85 MJ, which is 1.075 times higher than in the case without ammonia decomposition. However, to better understand the effect of pre-combustion ammonia reforming, hypothetically extract the two inlet energy streams (stream 1 and 2) from the outlet stream (stream 3). The first stream with an LHV of ammonia of 18.60 MJ provides $18.60 \times 0.884 = 16.44$ MJ of thermal exergy, which is 1.032 times higher than in the case without reforming. At the same time, the second heat stream of 4.99 MJ provides $4.99 \times 0.884 = 4.41$ MJ of thermal exergy, which is 1.275 times higher than in the case without reforming. Therefore, pre-combustion ammonia reforming increases the thermal exergy of the supplied external heat. Schematically, this thermal exergy transformation can be represented as shown in Fig. 3.

The example above illustrates that the pre-combustion ammonia decomposition increases the thermal exergy of two inlet streams and quantitative assessment of such an increase can be expressed via the coefficient of exergy transformation introduced above (11):

$$k_{ex,q(1)} = \frac{16.44\,\text{MJ/kg}}{15.93\,\text{MJ/kg}} = 1.032; \qquad k_{ex,q(2)} = \frac{4.41\,\text{MJ/kg}}{3.46\,\text{MJ/kg}} = 1.275. \tag{11}$$

Thus, the use of external heat at a temperature of 700 °C for pre-combustion ammonia decomposition increases the thermal exergy of this

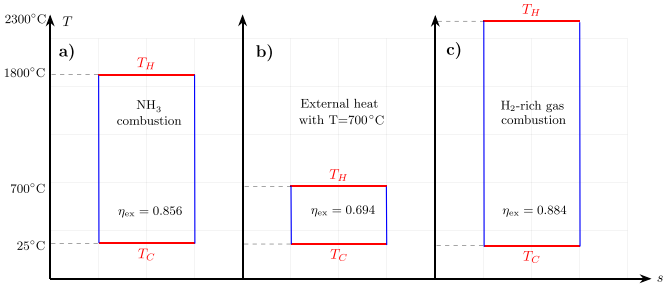


Fig. 3. Carnot cycles for thermal exergy transformation of the following energy streams: a) the first stream – heat with combustion products after ammonia combustion; b) the second stream – external heat at T = 700 °C; c) the third stream – heat with combustion products after H_2-rich gas combustion.

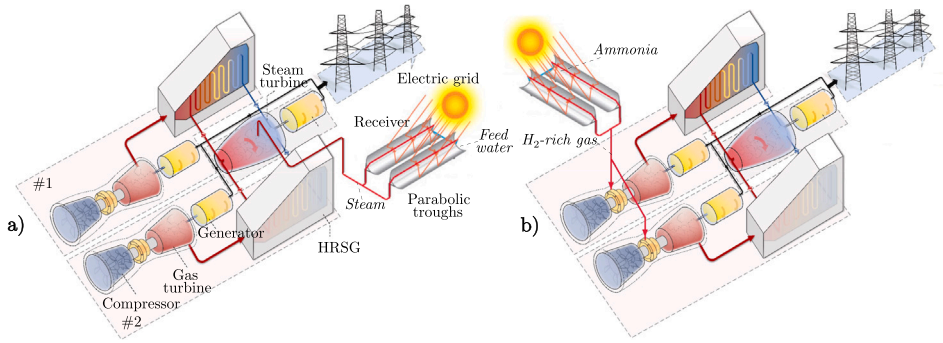


Fig. 4. Schematic diagrams of the two ammonia-fueled combined-cycle power plants integrated with concentrating solar thermal power: a) solar energy for steam generation; b) solar energy for pre-combustion ammonia decomposition.

heat stream by a factor of 1.275. This example demonstrates that thermal exergy can be enhanced, and pre-combustion ammonia decomposition can therefore be regarded as an “exergy pump”.

The fact that the coefficient of exergy transformation exceeds 1.0 does not violate the laws of thermodynamics, as it accounts not for the total exergy of the initial fuel, but rather for the increase in exergy attributable to the supplied heat. The initial exergy of ammonia (expressed in a form of LHV) serves as the baseline level in the system balance. The“exergy pump” effectively utilizes medium-potential heat to elevate this baseline. Conceptually, the operating principle of an “exergy pump” can be compared to that of a heat pump, which can transfer more thermal energy than the work expended. However, unlike a heat pump, an “exergy pump” receives a heat stream with lower exergy and delivers an outlet heat stream with higher exergy.

In the example above, the adiabatic flame temperatures were calculated assuming stoichiometric combustion ($\lambda = 1.0$) at atmospheric pressure (1 bar) with air (21% O_2, 79% N_2). Equilibrium compositions were obtained via Gibbs free energy minimization using Aspen HYSYS (Peng–Robinson equation of state). For ammonia, the calculated flame temperature of 1800 °C agrees with literature values [31,32]. For the H_2-rich gas (75% H_2, 25% N_2), the resulting temperature is approximately 2300 °C. A ±50 °C variation in these temperatures would alter the exergy results by less than 2%, leaving the main conclusions unchanged.

3.3. Application in power plant

The preceding example demonstrated the theoretical feasibility of enhancing the exergy of a heat stream at 700 °C via ammonia pre-decomposition, using the Carnot cycle as an ideal reference. However, because the Carnot cycle represents the theoretical maximum for heat-to-work conversion, a more realistic evaluation is necessary. To address this, the concept of pre-combustion ammonia decomposition as an “exergy pump” was quantitatively assessed in the context of an ammonia-fueled combined-cycle power plant integrated with concentrating solar thermal power.

Concentrating solar power (CSP) plants offer a promising approach for generating electricity from solar thermal energy. Typically, parabolic troughs concentrate sunlight to produce steam, which then drives a steam turbine. Despite the potential to achieve temperatures exceeding 1000 °C, commercially available systems generally operate in the range of 400–500 °C [33,34]. This moderate temperature level limits the thermal efficiency of conventional CSP configurations, making the proposed exergy pumping concept a compelling alternative for upgrading low-to-medium temperature solar heat.

3.3.1. Thermodynamics analysis

To evaluate the practical potential of this concept, two integrated combined-cycle power plant configurations were analyzed. In the first configuration, solar thermal energy is used to generate additional steam, which is integrated into the bottoming cycle alongside steam produced in the heat recovery steam generator (HRSG). In the second configuration, solar heat is employed for the pre-combustion decomposition of ammonia, producing a hydrogen-rich fuel that is subsequently combusted in the gas turbine. Schematic diagrams of both configurations are presented in Fig. 4; the concept for the graphical representation of these diagrams was adapted from Lu et al. [35].

The combined-cycle power plant modeled in this study is based on two Mitsubishi H-25 gas turbines, each with a nominal output of 40 MW, as reported in recent demonstrations of ammonia-fired gas turbine technology [36].

Thermodynamic analysis of the two configurations was performed in Aspen HYSYS. The main parameters of the combined cycle plant were selected according to the manufacturer’s recommendations. In the combined cycle plant based on the Mitsubishi H-25 gas turbine, the temperature of the working gases at the gas turbine inlet and the exhaust gas temperature are 1300 and 570 °C, respectively. The parameters of the steam generated in the heat recovery steam generator and solar receiver for the steam turbine in the base schemes (Fig. 4(a)) are 40 bar and 350 °C. In the calculation scheme, it is assumed that the temperature in the solar collector is 500 °C. The solar heat input (ΔH_{sol}) is identical for both configurations. In the configuration shown in Fig. 5(a), this thermal energy is used for steam generation, whereas in Fig. 5(b), it is employed for the pre-combustion decomposition of ammonia.

3.3.2. Model verification

Aspen HYSYS is widely used for the thermodynamic analysis of power generation cycles, with the literature demonstrating good agreement between simulation results and experimental data [37]. As a powerful steady-state process simulator, it is particularly well-suited for modeling endothermic chemical processes such as ammonia decomposition [27]. To ensure the reliability of the present simulations, a comprehensive verification procedure was implemented.

Since experimental data for a full-scale ammonia-fired CCPP are not yet available, verification was performed in two complementary steps. First, the thermodynamic model of the gas turbine cycle was validated by comparing its performance against an equivalent model built in GateCycle, a widely adopted commercial software in the power industry. In addition, the gas turbine model was verified using available data on fuel consumption for the H-25 gas turbine fueled with natural gas, hydrogen, and ammonia, as well as other performance metrics for ammonia-fired gas turbines [38].

Second, the ammonia decomposition sub-model was verified by comparing its predicted equilibrium composition against established experimental data from the literature [39].

Cycle performance verification

Table 1 compares the simulated performance of the H-25 gas turbine obtained from Aspen HYSYS and GateCycle against reference

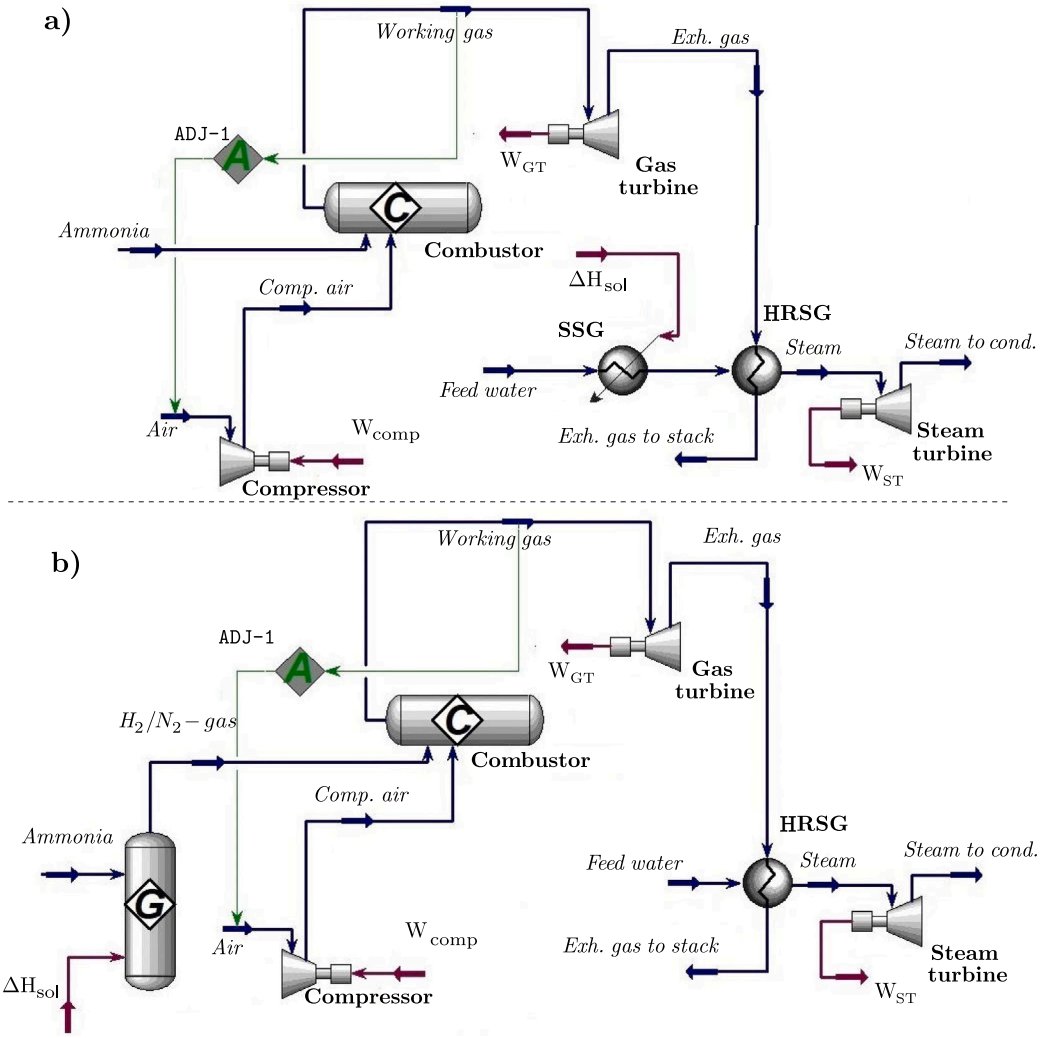


Fig. 5. Calculation scheme in the Aspen HYSYS software product for thermodynamic analysis of combined cycle plants using solar energy: (a) for thermochemical conversion of ammonia; (b) for steam generation.

Table 1
Comparison of H-25 gas turbine performance parameters from simulation and manufacturer data.

Parameter	Aspen HYSYS	GateCycle	Mitsubishi [38]
NG consumption (kg/h)	9086	9071	9000
H_2 consumption (kg/h)	4014	4008	4000
NH_3 consumption (kg/h)	21,512	21,498	21,400
NH_3 metrics:			
LHV efficiency (%)	36.31	36.27	36.20
Exhaust flow (kg/s)	118.8	118.5	118.2
Exhaust gas temperature (°C)	563	567	569
LHV heat rate (kJ/kWh)	9902	9925	9949

data from Mitsubishi [38]. The simulations were conducted under the manufacturer-specified operating conditions: a pressure ratio of 17.9 and a turbine inlet temperature of 1260 °C.

The results in Table 1 demonstrate good agreement between the simulations and the reference data with deviation of less than 1%. The minor deviations observed, particularly in natural gas consumption, can be attributed to differences in the assumed natural gas composition, as the exact fuel specification was not provided by the manufacturer.

Ammonia decomposition model verification

The equilibrium composition predicted by the Aspen HYSYS model for the ammonia decomposition process was verified against experimental data from the Fixed Nitrogen Research Laboratory (F.N.R.L) [39]. Table 2 compares the ammonia molar fraction in the product gas across a wide range of temperatures (350–500 °C) and pressures (10–100 bar).

Table 2
Comparison of ammonia molar fraction in decomposition products between simulation and experimental data [39].

Temperature (°C)	Source	Pressure (bar)			
		10	30	50	100
350	Simulation	7.30	17.69	25.00	–
	Experiment	7.36	17.79	25.09	–
375	Simulation	5.19	13.31	19.41	30.91
	Experiment	5.24	13.34	19.45	30.96
400	Simulation	3.86	9.99	15.07	24.84
	Experiment	3.84	10.09	15.09	24.89
425	Simulation	2.80	7.55	11.66	20.14
	Experiment	2.79	7.60	11.69	20.22
450	Simulation	1.99	5.75	9.09	16.30
	Experiment	2.03	5.79	9.12	16.34
475	Simulation	1.61	4.51	7.11	13.01
	Experiment	1.62	4.53	7.13	12.98
500	Simulation	1.21	3.51	5.61	10.41
	Experiment	1.20	3.48	5.58	10.40

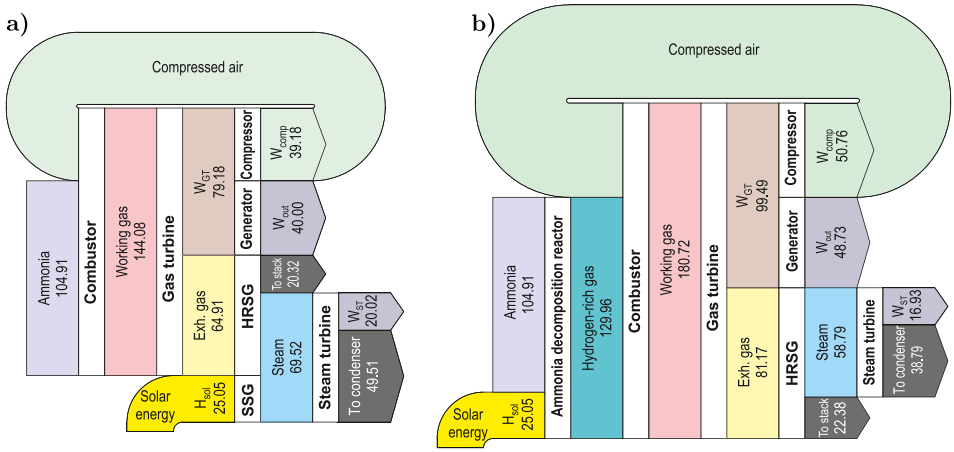


Fig. 6. Sankey diagrams of energy streams for the two configurations shown in Fig. 5 (numerical values in MW): a) solar energy ammonia decomposition; b) solar energy steam generation.

The simulated ammonia molar fractions show excellent agreement with the experimental data, with deviations consistently below 0.5% across all tested conditions. This confirms that the Aspen HYSYS model accurately captures the thermodynamic equilibrium of the ammonia decomposition reaction.

3.3.3. Energy balance

The energy balance of the two configurations illustrated in Fig. 5 is presented as Sankey diagrams in Fig. 6, depicting the conversion of chemical energy of ammonia and thermal solar energy into work. These diagrams correspond to the calculation scheme employed for thermodynamic analysis, with numerical values of each energy stream provided in MW.

The total input power for both configurations is 129.96 MW, consisting of 104.91 MW of chemical energy from ammonia and 25.05 MW of solar thermal energy. However, the manner in which solar energy is utilized fundamentally affects the distribution of work output between the gas turbine and steam turbine cycles.

When solar energy is used for the pre-combustion decomposition of ammonia (Fig. 6(a)), the gas turbine work output increases to 48.73 MW, compared to 40.00 MW in the nominal operating mode where ammonia is supplied directly to the combustion chamber. In contrast, when solar energy is used for additional steam generation in the bottoming cycle (Fig. 6(b)), the steam turbine work output increases to 20.02 MW, whereas steam generation using only exhaust gases from the gas turbine yields 16.93 MW.

Based on these energy balances, the thermal efficiency of the two configurations can be determined. The configuration employing solar energy for pre-combustion ammonia decomposition achieves an efficiency of 50.52%, while the configuration utilizing solar energy for steam generation yields an efficiency of 46.18%. This efficiency difference arises from the distinct work output distributions between the gas turbine and steam turbine cycles. In the ammonia decomposition configuration, the work output shares are 0.74 for the gas turbine and 0.26 for the steam turbine. In the steam generation configuration, these shares are 0.66 and 0.34, respectively.

Given that the thermal efficiency of the H-25 gas turbine cycle exceeds that of the steam turbine cycle for the specified parameters, increasing the proportion of work output from the gas turbine leads to more efficient utilization of medium-temperature solar heat at 500 °C. Consequently, from the perspective of the "exergy pump" concept, using solar thermal energy at 500 °C to drive pre-combustion ammonia decomposition is 23.9% more efficient than using it for steam generation in the bottoming cycle.

The efficiency values reported in this study (46.18% and 50.52%) are obtained from ideal thermodynamic simulations that do not account for practical losses such as reactor pressure drop, heat exchanger losses, solar receiver efficiency, catalyst bed thermal losses, or auxiliary power consumption. These factors would reduce the net efficiency gain in a real system. Nevertheless, the qualitative advantage of the "exergy pump" concept over conventional steam generation is expected to persist under realistic conditions, as both configurations share similar parasitic losses and the exergy upgrade mechanism provides a fundamental thermodynamic benefit.

3.4. Prospects for practical application

The practical implementation of the "exergy pump" concept based on pre-combustion fuel reforming depends critically on the appropriate selection of both the heat source and the energy carrier. The most accessible and technologically mature source of medium-temperature heat is concentrated solar energy and exhaust gas from gas turbine units, with modern systems achieving temperatures exceeding 500 °C. This opens the possibility of integrating thermochemical pre-combustion fuel reforming into existing combined cycles, utilizing heat that in conventional configurations is either partially lost or employed with low exergy efficiency in the steam power cycle [40,41].

Beyond ammonia, expanding the range of applicable fuels presents significant opportunities. Methanol (CH_3OH) emerges as a particularly promising candidate, especially when produced from renewable sources or with carbon capture, positioning it as a carbon-neutral fuel [42]. The principal advantage of methanol lies in its substantially lower conversion temperature, with complete decomposition achievable at approximately 300–350 °C [43]. This characteristic broadens the applicability of the "exergy pump" concept, enabling efficient utilization of heat streams with lower temperature potentials than those required for ammonia decomposition. Consequently, it facilitates the integration of additional low-temperature heat sources into the energy conversion cycle, enhancing overall system flexibility and efficiency.

A critical practical consideration for the proposed exergy pump concept is the combustion behavior of ammonia and hydrogen-rich mixtures in gas turbine combustors. Ammonia combustion inherently produces significant nitrogen oxides (NOx) due to fuel-bound nitrogen, while the hydrogen-rich product gas (up to 75% H_2) exhibits a higher flame speed and increased flashback tendency compared to natural gas or pure ammonia. These characteristics pose challenges for combustor stability, flame anchoring, and emissions compliance. Although the present thermodynamic analysis does not include detailed combustion chemistry or emissions modeling, it is recognized that practical implementation of

ammonia decomposition in gas turbines requires dedicated combustor designs and NOx abatement strategies. Promising solutions include two-stage rich-lean combustion, selective catalytic reduction (SCR), and micromix combustor concepts that have been successfully tested for ammonia and hydrogen fuels [23]. Future work should integrate such measures into the exergy pump framework to assess the overall system performance, including emissions and operability.

4. Conclusions

This study systematically investigated the thermochemical pre-combustion transformation of ammonia as an "exergy pump" for upgrading medium-temperature heat in power generation applications. The thermodynamic analysis confirmed that ammonia conversion exceeding 90% is achievable at temperatures above 600 °C, with the endothermic heat demand reaching 4.99 MJ/kg at 700 °C comprising both the reaction enthalpy of 3.01 MJ/kg and the sensible enthalpy increase of the reacting mixture of 1.98 MJ/kg.

The energy transformation analysis yielded coefficients of heat transformation of $k_{q(I)} = 1.162$ based on lower heating value and $k_{q(II)} = 1.268$ based on gross available heat. Furthermore, the exergy analysis demonstrated that pre-combustion ammonia decomposition increases the thermal exergy of the supplied heat stream by a factor of 1.275 at 700 °C, with coefficients of exergy transformation of $k_{ex,q(1)} = 1.032$ for the ammonia fuel stream and $k_{ex,q(2)} = 1.275$ for the external heat stream. This amplification represents the upgrading of low-quality thermal exergy to high-quality chemical exergy through endothermic reforming and does not violate thermodynamic laws.

In the ammonia-fueled combined-cycle power plant based on two Mitsubishi H-25 gas turbines with a total nominal power of 80 MW, the configuration utilizing solar thermal energy of 25.05 MW at 500 °C for pre-combustion ammonia decomposition achieved an overall thermal efficiency of 50.52%. This represents a 23.9% relative improvement over the configuration using the same solar input for steam generation in the bottoming cycle, which achieved 46.18% efficiency. The efficiency advantage stems from the increased work output share of the gas turbine of 0.74 versus 0.66 and the effective upgrading of medium-temperature solar heat via the "exergy pump" mechanism.

The "exergy pump" concept extends beyond ammonia to other endothermically reformable fuels, with methanol emerging as a particularly promising alternative due to its substantially lower decomposition temperature of 300–350 °C. This enables utilization of lower-temperature heat sources including industrial waste heat, geothermal energy, and lower-temperature solar thermal collectors. When produced from renewable sources or with carbon capture, methanol offers a carbon-neutral pathway compatible with the "exergy pump" framework.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (NSFC) under Grant No. W2532039. The author expresses gratitude to Mitsubishi Heavy Industries for providing reference data for thermodynamic analysis of the combined cycle plant schematic based on the Mitsubishi H-25 gas turbine.

Data availability

Data will be made available on request.

References

[1] Smith O, et al. The effect of renewable energy incorporation on power grid stability and resilience. Sci Adv 2022;8(9):eabj6734.
[2] Schmietendorf K, Peinke J, Kamps O. The impact of turbulent renewable energy production on power grid stability and quality. Eur Phys J B 2017;90(11):222.
[3] Giampaolo T. Gas turbine handbook: principles and practice. River Publishers; 2020.
[4] Zoran Rant. Exergie, ein neues wort für technische arbeitsfähigkeit. In: Forschung auf DEM gebiete des ingenieurwesens, vol. 22; 1956. p. 36–7.
[5] Michaelides E. Exergy analysis for energy conversion systems. Cambridge University Press; 2021.
[6] Carnot S. Réflexions sur la puissance motrice du feu et sur les machines propres à développer cette puissance. Ann sci École norm supérieure 1872;1:393–457.
[7] Clausius R. Ueber verschiedene für die anwendung bequeme formen der hauptgleichungen der mechanischen wärmetheorie:(vorgetragen in der naturforsch. In: Gesellschaft den, vol. 24. Verlag Nicht Ermittelbar. Apr 1865.
[8] Rostrup-Nielsen JR. Steam reforming and chemical recuperation. Catal Today 2009;145(1–2):72–5.
[9] Tartakovsky L, Sheintuch M. Fuel reforming in internal combustion engines. Prog Energy Combust Sci 2018;67:88–114.
[10] Popov SK, et al. The use of thermochemical recuperation in an industrial plant. Energy 2017;127:44–51.
[11] Pashchenko D. First law energy analysis of thermochemical waste-heat recuperation by steam methane reforming. Energy 2018;143:478–87.
[12] Pashchenko D. Integrated solar combined cycle system with steam methane reforming: thermodynamic analysis. Int J Hydrog Energy 2023;48(48):18166–76.
[13] Zamfirescu C, Dincer I. Using ammonia as a sustainable fuel. J Power Sources 2008;185(1):459–65.
[14] Pashchenko DI. Ammonia as a fuel for gas-turbine units with thermochemical recuperation of exhaust gas heat. Therm Eng 2025;72(1):1–7.
[15] Pashchenko D. Ammonia fired gas turbines: recent advances and future perspectives. Energy 2024;290:130275.
[16] Valera-Medina A, et al. Ammonia for power. Prog Energy Combust Sci 2018;69:63–102.
[17] Figueroa-Labastida M, et al. Ammonia/hydrogen laminar flame speed measurements at elevated temperatures. Int J Hydrog Energy 2024;63:1137–46.
[18] Chai WS, et al. A review on ammonia, ammonia-hydrogen and ammonia-methane fuels. Renew Sustain Energy Rev 2021;147:111254.
[19] Pashchenko D. Ammonia-fueled solar combined cycle power system: thermal and combustion efficiency enhancement via solar-driven ammonia decomposition. Int J Hydrog Energy 2025;142:232–43.
[20] Pashchenko D. Natural gas reforming in thermochemical waste-heat recuperation systems: a review. Energy 2022;251:123854.
[21] Pashchenko D. Thermochemical waste-heat recuperation as on-board hydrogen production technology. Int J Hydrog Energy 2021;46(57):28961–8.
[22] Herbinet O, Bartocci P, Dana AG. On the use of ammonia as a fuel–a perspective. Fuel Commun 2022;11:100064.
[23] Valera-Medina A, et al. Review on ammonia as a potential fuel: from synthesis to economics. Energy Fuels 2021;35(9):6964–7029.
[24] Dimitriou P, Javaid R. A review of ammonia as a compression ignition engine fuel. Int J Hydrog Energy 2020;45(11):7098–118.
[25] Lamb KE, Dolan MD, Kennedy DF. Ammonia for hydrogen storage; a review of catalytic ammonia decomposition and hydrogen separation and purification. Int J Hydrog Energy 2019;44(7):3580–93.
[26] Schüth F, et al. Ammonia as a possible element in an energy infrastructure: catalysts for ammonia decomposition. Energy Environ Sci 2012;5(4):6278–89.
[27] Pashchenko D, Mustafin R, Karpilov I. Ammonia-fired chemically recuperated gas turbine: thermodynamic analysis of cycle and recuperation system. Energy 2022;252:124081.
[28] Pashchenko D. Thermodynamic equilibrium analysis of combined dry and steam reforming of propane for thermochemical waste-heat recuperation. Int J Hydrog Energy 2017;42(22):14926–35.
[29] Lucentini I, et al. Review of the decomposition of ammonia to generate hydrogen. Ind Eng Chem Res 2021;60(51):18560–611.
[30] Lucentini I, et al. Catalytic ammonia decomposition over ni-ru supported on CeO2 for hydrogen production: effect of metal loading and kinetic analysis. Appl Catal B Environ 2021;286:119896.
[31] Duynslaegher C, Jeanmart H, Vandooren J. Ammonia combustion at elevated pressure and temperature conditions. Fuel 2010;89(11):3540–5.
[32] Han X, et al. The temperature dependence of the laminar burning velocity and superadiabatic flame temperature phenomenon for NH_3/air flames. Combust Flame 2020;217:314–20.
[33] Zhang HL, et al. Concentrated solar power plants: review and design methodology. Renew Sustain Energy Rev 2013;22:466–81.
[34] Yao L, Wang Y, Xiao X. Concentrated solar power plant modeling for power system studies. IEEE Trans Power Syst 2023;39(2):4252–63.
[35] Nianci L, et al. A two-dimensional design and synthesis method for coordinated control of flexible-operational combined cycle of gas turbine. Energy 2023;284:128576.

[36] Nose M, et al. Development of hydrogen/ammonia firing gas turbine for decarbonized society. Mitsubishi Heavy Ind Tech Rev 2021;58(1).
[37] Liu Z, Karimi IA. Simulating combined cycle gas turbine power plants in aspen HYSYS. Energy Convers Manag 2018;171:1213–25.
[38] Mitsubishi Power. Hydrogen Power generation handbook (fifth edition): towards the realization of integrated hydrogen technologies from production to power generation. Fifth edition. Includes technical papers and updates on hydrogen energy trends, 2023.
[39] Larson AT, Dodge RL. The ammonia equilibrium. J Am Chem Soc 1923;45(12):2918–30.
[40] Pashchenko D. Performance evaluation of a combined power generation system integrated with thermochemical exhaust heat recuperation based on steam methane reforming. Int J Hydrog Energy 2023;48(15):5823–35.
[41] Pashchenko D. Hydrogen-enriched combustion via exhaust-driven ammonia reforming: a pathway to high-efficiency gas turbine operation. Therm Sci Eng Prog 2025;104229.
[42] Deka TJ, et al. Methanol fuel production, utilization, and techno-economy: a review. Environ Chem Lett 2022;20(6):3525–54.
[43] Pashchenko D. Low-grade heat utilization in the methanol-fired gas turbines through a thermochemical fuel transformation. Therm Sci Eng Prog 2022;36:101537.